

What kills patients with extraskkeletal myxoid chondrosarcoma, and the survival available to tumour-directed therapy: a cause-of-death and relative-survival analysis of the published record

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What kills patients with extraskkeletal myxoid chondrosarcoma, and the survival available to tumour-directed therapy: a cause-of-death and relative-survival analysis of the published record

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A study of the published record. No patient was recruited, no patient-level record was re-analysed, and no claim is made that any treatment works. Every figure below is computed from counts and survival percentages already published, or from a public national life table. Analyses were carried out with AI assistance (see Methods).

Abstract

Background. Extraskkeletal myxoid chondrosarcoma is an ultra-rare translocation sarcoma with a protracted course: reported ten-year disease-specific survival is approximately 85 per cent, while 30 to 50 per cent of patients develop distant metastases, mostly pulmonary. Research effort in this disease is directed almost entirely at preventing death from the tumour. How often patients die of the tumour, and by what mechanism, has not been described.

Methods. Six hundred open-access publications matching the disease were enumerated through Europe PMC and 328 full texts retrieved. Sentences containing a death cue were extracted mechanically and classified by reading, with the patient rather than the sentence as the

unit; papers not about this disease were excluded, leaving 34. Cause-of-death splits reported on identical patients were extracted from a national registry cohort. Relative survival was computed under the Ederer II convention against a World Health Organization life table for the United States, matched to the cohort median age and sex ratio. Every quoted figure is asserted against its source artifact programmatically, and the analysis code and data are public.

Results. Of 52 deaths described in the disease's open-access literature, 15 (28.8 per cent) carried any stated mechanism. Among those 15, death from a competing cause or a second malignancy was the largest category (6 deaths, 40 per cent), exceeding respiratory failure (3 deaths), and one of the three respiratory deaths followed tumour-embolic ischaemic stroke rather than pulmonary tumour burden. In the national registry cohort, deaths not attributed to the sarcoma comprised 30.8 per cent of deaths in localised disease (4 of 13) and 10.0 per cent in disease metastatic at diagnosis (1 of 10). The corresponding upper bound on what preventing every disease death could add was 6.7 percentage points of survival in localised disease and 31.0 points in metastatic disease at a median follow-up of just over three years. Relative survival, which assigns no cause to any patient, gave a median competing share of 23.0 per cent across eight series-horizons, against 21.7 per cent from the cause split; the two methods share no input. Observed non-sarcoma deaths matched background mortality for the cohort's age and sex (ratios 0.97 and 1.04, on four and one events). No publication reporting the growth rate of pulmonary metastases in this disease was found.

Conclusions. The published record of this disease cannot say how most of its patients die. What it does say is that a substantial minority of deaths are not caused by the sarcoma, that this fraction is large in localised disease and small in metastatic disease, and that the survival available to antitumour therapy is correspondingly concentrated in the metastatic stratum. Two independent methods agree on the size of the competing fraction. These findings bound what tumour-directed research in this disease can achieve and identify a second, unaddressed population of deaths.

1. Introduction

Extraskelatal myxoid chondrosarcoma is defined by rearrangement of *NR4A3*, most often as an *EWSR1::NR4A3* fusion. It is ultra-rare, with an incidence well under one per million per year, and its course is unusually protracted: patients recur and metastasise late, and survival with metastatic disease is measured in years rather than months.

Almost all translational work in this disease is directed at the tumour. That is reasonable, and it carries an assumption nobody has examined: that the event research is trying to prevent, death caused by the sarcoma, is the event that actually occurs. For a disease whose reported ten-year disease-specific survival is approximately 85 per cent, in patients most often diagnosed in their fifties and sixties, the assumption is not self-evident. If a meaningful share of deaths after diagnosis are not caused by the sarcoma, then the survival available to any antitumour therapy is smaller than overall survival curves suggest, and a second population of deaths exists that no tumour-directed approach addresses.

Answering this requires knowing what patients die of, which requires the literature to have recorded it. This study asks three questions in order: how often the published record states a mechanism of death; what fraction of deaths are attributed to the disease when a cohort reports both; and whether a method that requires no cause-of-death assignment agrees.

2. Methods

2.1 Corpus

Publications were enumerated through the Europe PMC REST interface using a disease query spanning the condition's alternative names, restricted to records with open-access or Europe PMC full text. Six hundred records were enumerated and full text retrieved for 400, of which 328 returned parsable text. Reference lists, tables and figures were removed before analysis, because bibliographies contain other papers' titles carrying death terms.

Sentences containing a death cue were extracted by pattern. The pattern is recall-oriented by design: it retains negative statements and methodological definitions, because a filter tuned to exclude them would also exclude unusual terminal events, which are the observations of interest. No cause was assigned by pattern at any point.

2.2 Inclusion and classification

Of 162 papers carrying at least one death sentence, 34 have the disease in their title. The remainder match because the disease appears in a differential diagnosis or a citation, and their deaths belong to other diseases' patients; all analyses are restricted to the 34.

Classification was performed by reading each sentence with its paper's context, and assigning one of: respiratory failure, locoregional complication, visceral metastasis complication, treatment-related death, competing non-cancer death, second malignancy, death from progressive disease without a stated mechanism, mechanism unstated, or not a patient death.

The unit of analysis is a patient or a reported patient group, never a sentence. One paper describes three deaths across seven sentences and another describes one death in four; counting sentences would have inflated the total roughly threefold and weighted the most verbose reports most heavily.

2.3 Cause splits and the upper bound on antitumour benefit

Where a cohort reported both deaths attributed to the sarcoma and deaths attributed to other causes on identical patients, the competing share was computed as a direct ratio of counts. This requires no survival curve, no pairing across populations and no subtraction of summary percentages.

The upper bound on antitumour benefit, referred to below as the ceiling, is the disease-specific mortality present in a cohort: the survival that preventing every death from the sarcoma would add. It is an upper bound on all tumour-directed therapy taken together, not on any single agent. Binomial proportions carry Wilson intervals.

2.4 Relative survival

Relative survival was computed as observed all-cause survival divided by the survival expected for a general population of the same age and sex, under the Ederer II convention, using age-specific death rates from the World Health Organization Global Health Observatory for the United States. Expected survival was computed for a synthetic cohort at the median age at diagnosis and the reported sex ratio, and survival across a rate band of width w at rate m was taken as $\exp(-mw)$.

This method attributes excess mortality to the disease without assigning a cause to any individual, which is the point: it is independent of the cause-split described above and shares none of its inputs. Series with relative survival at or above 0.99 were excluded from the pooled band, because the competing share then divides a near-zero excess by a small all-cause mortality and becomes numerically unstable; exclusions are reported individually.

2.5 Reproducibility and the use of AI assistance

All analysis code, input specifications and output artifacts are public. Every survival figure is declared with the verbatim string it was read from and checked programmatically against its source, so a corrected source figure fails the build rather than leaving a stale number. Every classified death carries its verbatim sentence and identifier, checked against the retrieval artifact before any tally is produced.

Analyses were carried out with AI assistance. Retrieval, arithmetic and consistency checking were performed by software; the classification of each death, the inclusion decisions and the interpretation are the author's. Two errors found during the work are recorded in Appendix A, because both changed a headline figure.

3. Results

3.1 Statement of a mechanism in the published record

Fifty-two deaths were described across 16 of the 34 papers. Fifteen (28.8 per cent) carried a stated mechanism. Twenty-eight were recorded with no mechanism beyond a vital status, and a further eight were attributed to progressive disease without further specification.

Table 1. Deaths in the open-access literature of this disease, by stated mechanism.

category	deaths
mechanism unstated	28
progressive disease, mechanism unspecified	8
respiratory failure	3

competing non-cancer cause	3
second malignancy	3
locoregional complication	2
visceral metastasis complication	2
treatment-related	2
ambiguous	1
total	52

This is the study's first result rather than a limitation of it. A research programme directed at preventing a specific event cannot, from its own literature, describe that event in most of the patients it has recorded.

3.2 Competing causes and second malignancies

Among the 15 deaths with a stated mechanism, six were from a competing cause or a second malignancy, exceeding any disease-specific mechanism. These recur across independent reports: two patients died of concurrent malignancies within months of diagnosis in one series; a patient followed for 126 months with metastatic disease died of unresectable colon cancer; two of sixteen collected intracranial cases died of causes the reviewing authors judged unrelated; and one patient died of cerebral haemorrhage at 9.5 years with no tumour recurrence.

Case reports over-select the notable, so this is a description of what was recorded and not an incidence. Its direction is nevertheless the same as the registry result below, which is not subject to case-report selection.

3.3 Respiratory failure, and the distinction from metastatic site

Three deaths were attributed to respiratory failure or to lung metastases. One followed tumour-embolic ischaemic stroke treated by thrombectomy, with death 15 days later, rather than progressive pulmonary tumour burden.

This distinction matters because the pulmonary dominance of this disease's metastatic pattern is well established and is easily transposed into an assumption about how patients die. The site of metastasis and the mode of death are different claims, and the present corpus supports the first without establishing the second.

3.4 The antitumour ceiling, and its dependence on stage

One national registry cohort reported both cause counts on identical patients.

Table 2. Cause of death and the ceiling on antitumour benefit, by stage at diagnosis.

stratum	n	disease deaths	other-cause deaths	competing share	ceiling
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localised, surgically treated	134	9	4	30.8 %	6.7 points
metastatic at diagnosis	29	9	1	10.0 %	31.0 points
combined	163	18	5	21.7 %	11.0 points

At a median follow-up of just over three years, preventing every death from the sarcoma in the localised cohort would have added 6.7 percentage points of survival; in the cohort metastatic at diagnosis it would have added 31.0. The value available to tumour-directed therapy is concentrated in the stratum that is the minority of patients.

The same cohort reported that distant metastasis was associated with death from the tumour on multivariate analysis, with an odds ratio of 26.26 (95 per cent CI 2.99 to 231.04), while local recurrence was not associated with tumour mortality. The confidence interval is very wide on a small number of events.

3.5 Agreement between two independent estimates

Relative survival gave a median competing share of 23.0 per cent across eight series-horizons (range 12.1 to 45.4), against 21.7 per cent from the registry cause split. The two share no input: one is published all-cause survival divided by a national life table, the other is counts of patients a registry assigned a cause to, and neither can be derived from the other.

One series was excluded from the band with relative survival of 0.9925, and one horizon of the same series produced relative survival above 1, meaning the cohort out-lived the general population. That is a selection signal in a 42-patient cohort reporting 100 per cent five-year survival, not a property of the disease.

3.6 Non-sarcoma deaths occur at background rate

Observed deaths not attributed to the sarcoma were compared against a life table matched on age and sex, at each cohort's own follow-up: 3.0 per cent observed against 3.1 per cent expected in the localised stratum (ratio 0.97, 95 per cent CI 0.38 to 2.41) and 3.4 against 3.3 per cent in the metastatic stratum (ratio 1.04, 95 per cent CI 0.18 to 5.18).

These rest on four events and one event respectively. What they establish is consistency with background mortality, not equality to it. The comparison is also one-sided: a general-population life table overstates background mortality for a cohort fit enough to reach and survive a sarcoma diagnosis, so it can refute the proposition that the gap is ordinary background mortality and cannot prove it. It did not refute it.

3.7 Absence of a growth-rate measurement for pulmonary metastases

A targeted search for the natural history, growth rate or doubling time of pulmonary metastases in this disease returned zero records. Whether repeated local treatment of lung metastases can keep pace with their progression depends on that rate, and it has not been measured.

4. Discussion

4.1 What this bounds

Preventing every death caused by the sarcoma in the localised cohort would have added 6.7 percentage points of survival at three years. This is an upper bound on all tumour-directed therapy taken together, and no agent achieves an upper bound. It does not argue against developing such therapy: in metastatic disease the same calculation gives 31.0 points, and that is where the value is.

What it does argue against is a portfolio that regards all stages as one target. The stratum with most patients has least to gain, and the stratum with most to gain is a minority.

4.2 What this says about the evidence base

That fewer than a third of recorded deaths carry a stated mechanism is a finding about the literature rather than the disease. It has a practical consequence: every disease-specific survival figure in this disease rests on somebody having assigned a cause by an instrument no paper reports. The relative survival analysis was performed because it does not need that assignment, and its agreement with the cause split provides evidence that the assignment, where made, is not badly wrong.

For ultra-rare cancers generally, the implication is that cause-of-death recording, which costs nothing at the point of follow-up, determines whether a disease's evidence base can support the questions its research programme asks.

4.3 The population no tumour-directed approach addresses

Between a fifth and a third of deaths after diagnosis are not caused by the sarcoma, and they occur at approximately the rate the cohort's age and sex predict. Two consequences follow, and only the first is established here.

Established: the ceiling on antitumour benefit is correspondingly reduced, most severely in localised disease.

Not established, and stated as a direction rather than a result: if these deaths occur at general-population rates, then interventions that reduce general-population mortality would be expected to apply, with a weaker transportability assumption than any tumour-directed therapy requires, because the deaths in question are not cancer deaths. Quantifying that requires effect sizes this study did not retrieve, and no outcome data on supportive or host-directed care exists in this disease in either direction.

4.4 Lung-directed treatment

Pulmonary metastasis is the dominant distant pattern in this disease, and both surgical metastasectomy and ablative local therapies are established options in sarcoma more broadly. The present corpus does not, however, establish respiratory failure as the dominant mode of death, so this study cannot support prioritising lung-directed treatment on mortality-mechanism grounds. The absence of any growth-rate measurement (3.7) is the more immediate obstacle: whether repeated local therapy can outpace this disease's pulmonary progression is unknown, and it is answerable from serial imaging already held by treating centres.

5. Limitations

Open-access full text is a convenience sample: 328 of 600 enumerated records were retrieved, and non-open-access series are systematically older and larger. Case reports are written because a case was notable, over-representing unusual terminal events; this biases against the indolent, competing-cause picture the data nevertheless show. Counts of described patients have no denominator and support no rate.

The cause split rests on 13 and 10 deaths in two strata of one registry, at a median follow-up of just over three years, which is short relative to this disease's natural history. Follow-up censors the two causes unequally: competing deaths accrue immediately while sarcoma deaths continue to accrue for decades, so short follow-up flatters the competing share. The ceiling figures are therefore lower bounds on the eventual disease mortality and the competing share is an upper bound at that horizon.

Relative survival was applied to published summary survival with a cohort median age and sex ratio rather than patient-level data, so the expected-survival term is that of a synthetic cohort rather than a true Ederer II match. A single national life table for one country and year was applied to cohorts from five countries across five decades. Excess mortality attributes to the disease any elevation in other-cause mortality among its patients, including treatment-caused death, which the cause split counts separately.

Where a paper judged a death unrelated to the sarcoma, that is the paper's judgement, made by an instrument it does not report.

6. Conclusion

The published record of extraskeletal myxoid chondrosarcoma does not say how most of its patients die. Where it does, deaths from competing causes and second malignancies are the largest identifiable category, and respiratory failure, though present, is not dominant. Between a fifth and a third of deaths after diagnosis are not caused by the sarcoma, a figure two independent methods agree on, and those deaths occur at approximately background rate. The survival available to antitumour therapy is 6.7 percentage points in localised disease and 31.0 in metastatic disease at three years. Research prioritisation in this disease should reflect that difference, and cause-of-death recording should be treated as a measurement that determines what its evidence base can answer.

Appendix A. Corrections made during this work

Both corrections changed a headline figure and are recorded because the superseded values were circulated internally before they were withdrawn.

A.1 Competing share, 39.4 per cent superseded by 21.7 per cent. The first estimate paired a survival-curve reading against a crude death proportion from a different cohort. When one registry's own cause counts on identical patients became available, the direct ratio replaced it. The direct estimator has none of the weaknesses the first disclosed, and the change is a change of method rather than a correction of arithmetic.

A.2 Background mortality, 2.4 per cent superseded by 11.3 per cent at ten years. The life-table fetch treated the World Health Organization indicator nM_x as a five-year probability. It is an age-specific death rate, and survival over a band of width w is $\exp(-mw)$. The error understated background mortality roughly fourfold and would have attributed nearly every death to the sarcoma. It was caught because the resulting figure was implausibly low for a cohort at this age, and confirmed from the indicator's published name rather than by inspection of which value looked right.

Data and code availability

All analysis code, input specifications, retrieval artifacts and outputs are available in the project repository. The retrieval artifacts record every query verbatim with its hit count, so both positive findings and absences can be reproduced or refuted.